

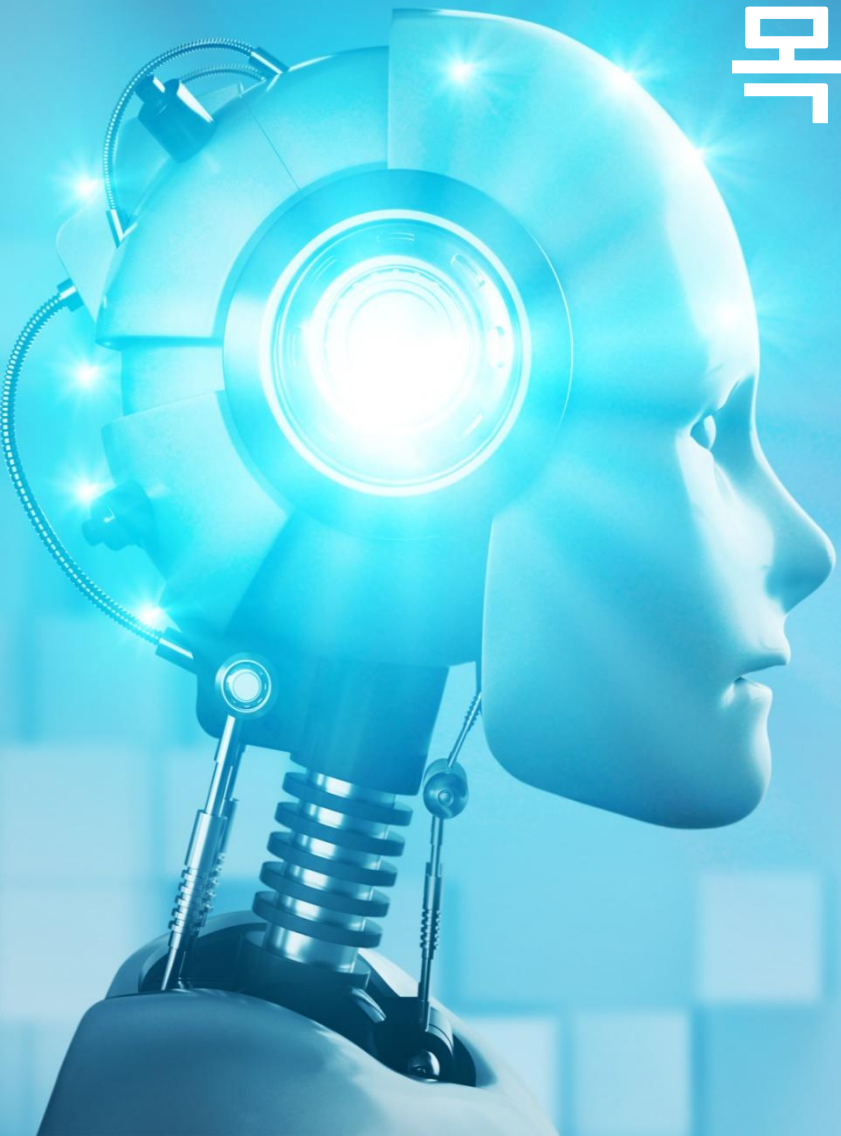
인공지능(AI) 개발자 양성 과정

별첨

DBMS 환경 설정

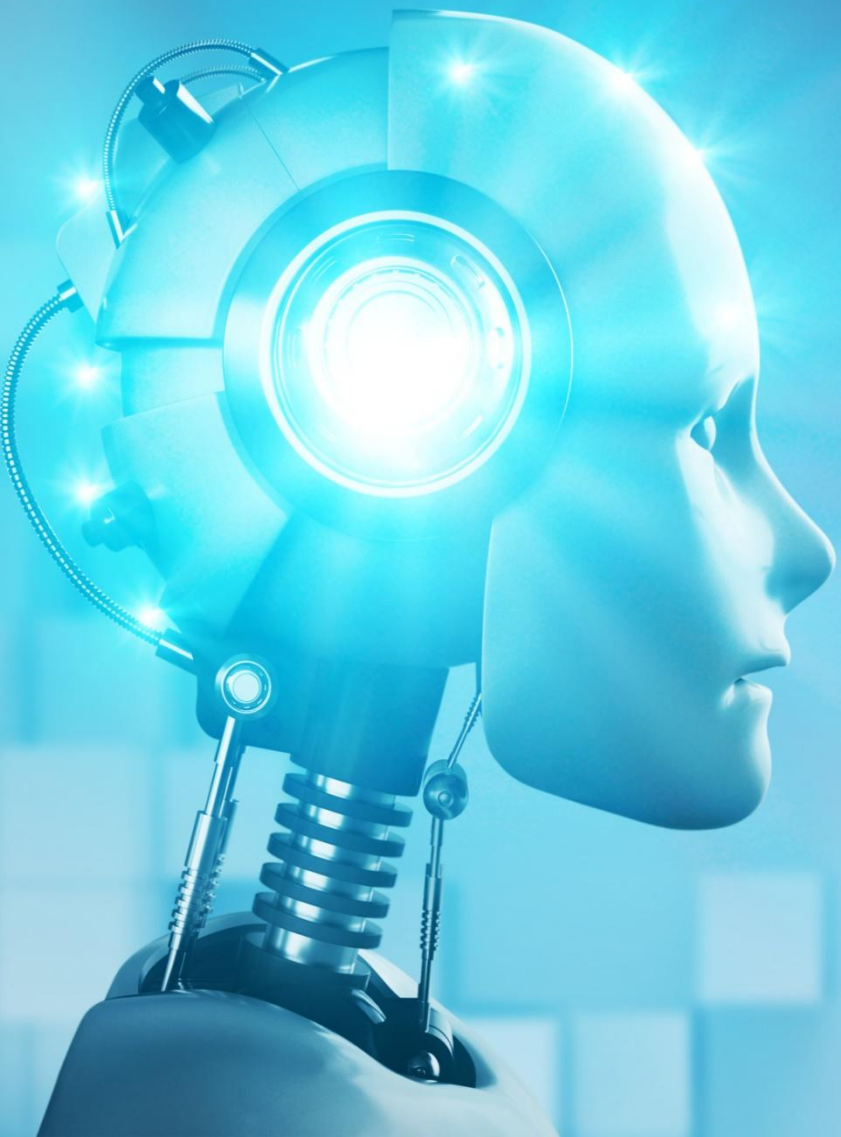
휴먼교육센터 안홍근 강사





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1. mariadb 설치
2. dbeaver 설치



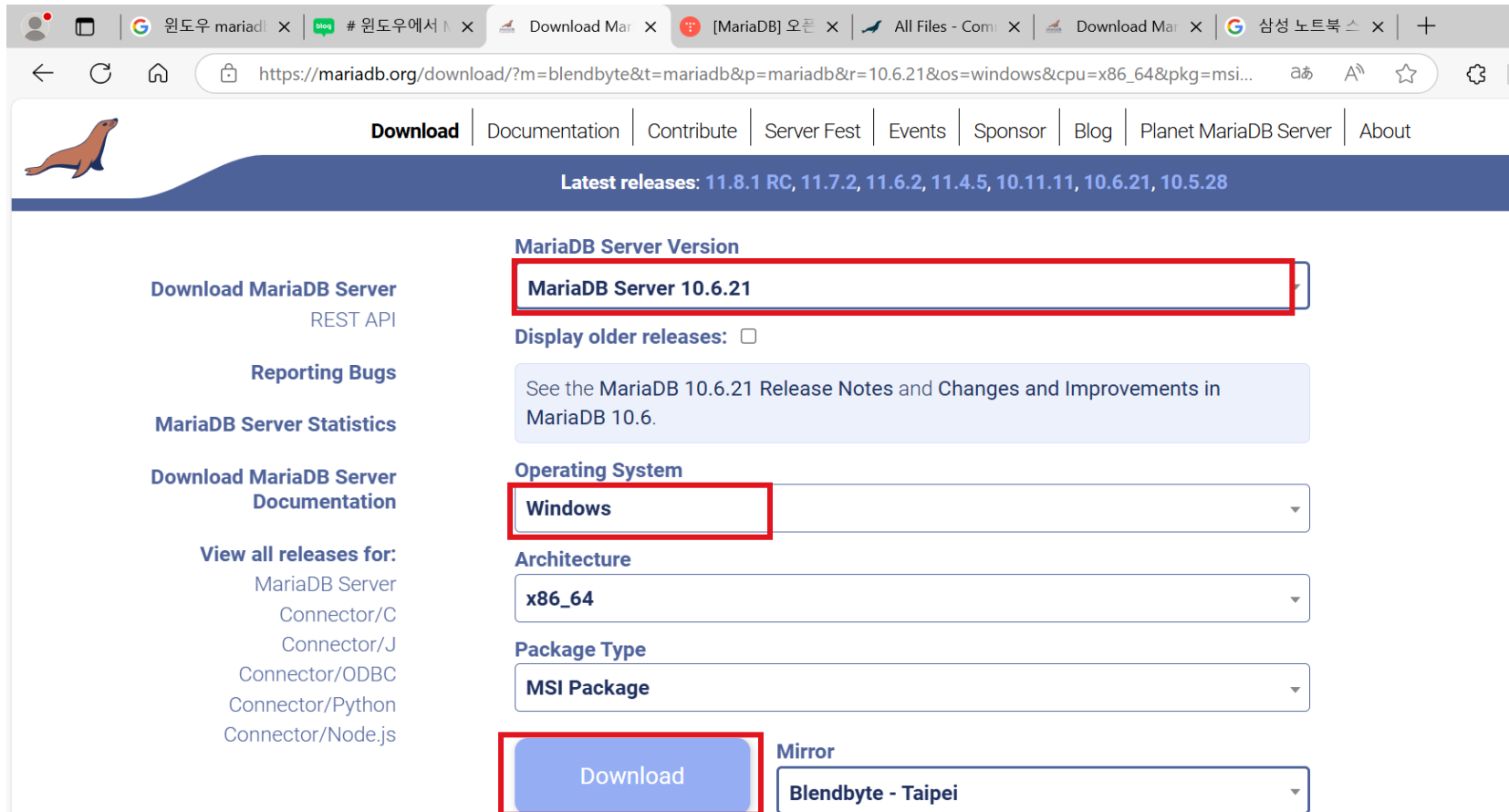
1 Mariadb 설치

1. Mariadb 설치

1. Mariadb 설치

☑ 설치 URL : <https://mariadb.org/download>

☑ Mariadb 버전 : 10.6.21 설치, 다운로드 실행



The screenshot shows the MariaDB download page. The browser's address bar displays the URL: https://mariadb.org/download/?m=blendbyte&t=mariadb&p=mariadb&r=10.6.21&os=windows&cpu=x86_64&pkg=msi.... The page features a navigation bar with links: Download, Documentation, Contribute, Server Fest, Events, Sponsor, Blog, Planet MariaDB Server, and About. A banner below the navigation bar lists the latest releases: 11.8.1 RC, 11.7.2, 11.6.2, 11.4.5, 10.11.11, 10.6.21, and 10.5.28. On the left side, there are links for Download MariaDB Server, REST API, Reporting Bugs, MariaDB Server Statistics, Download MariaDB Server Documentation, and View all releases for: MariaDB Server, Connector/C, Connector/J, Connector/ODBC, Connector/Python, and Connector/Node.js. The main content area includes a 'MariaDB Server Version' dropdown menu set to 'MariaDB Server 10.6.21', a 'Display older releases' checkbox, a text box with release notes for MariaDB 10.6.21, an 'Operating System' dropdown menu set to 'Windows', an 'Architecture' dropdown menu set to 'x86_64', a 'Package Type' dropdown menu set to 'MSI Package', a blue 'Download' button, and a 'Mirror' dropdown menu set to 'Blendbyte - Taipei'.

Download | Documentation | Contribute | Server Fest | Events | Sponsor | Blog | Planet MariaDB Server | About

Latest releases: 11.8.1 RC, 11.7.2, 11.6.2, 11.4.5, 10.11.11, 10.6.21, 10.5.28

Download MariaDB Server
REST API

Reporting Bugs

MariaDB Server Statistics

Download MariaDB Server
Documentation

View all releases for:
MariaDB Server
Connector/C
Connector/J
Connector/ODBC
Connector/Python
Connector/Node.js

MariaDB Server Version
MariaDB Server 10.6.21

Display older releases: ☐

See the MariaDB 10.6.21 Release Notes and Changes and Improvements in MariaDB 10.6.

Operating System
Windows

Architecture
x86_64

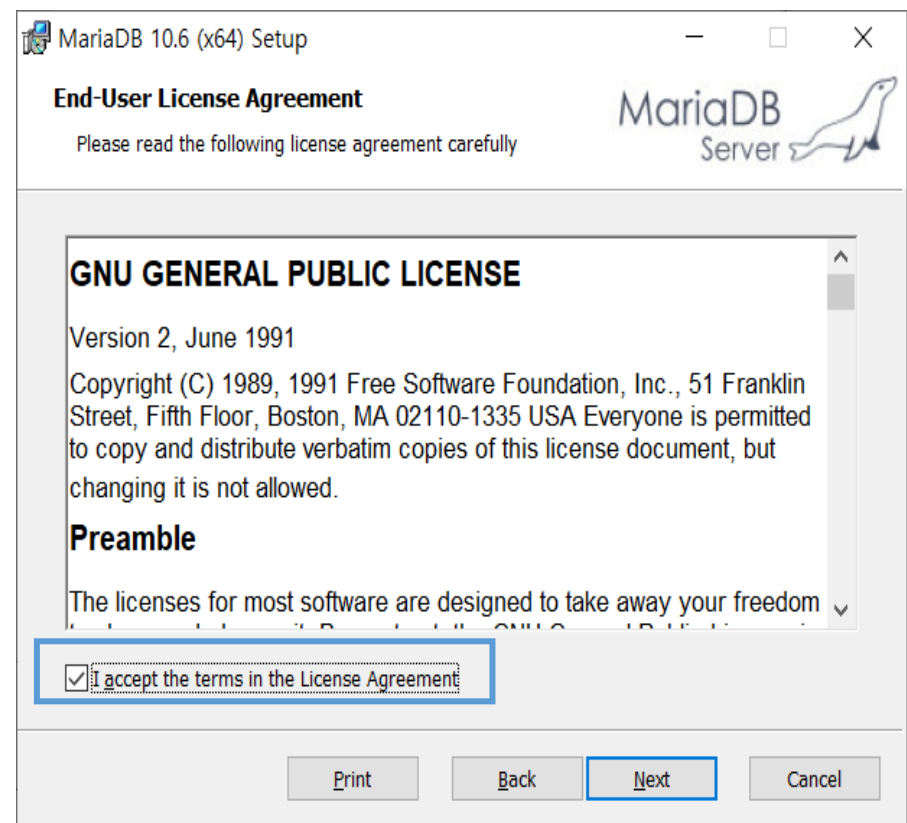
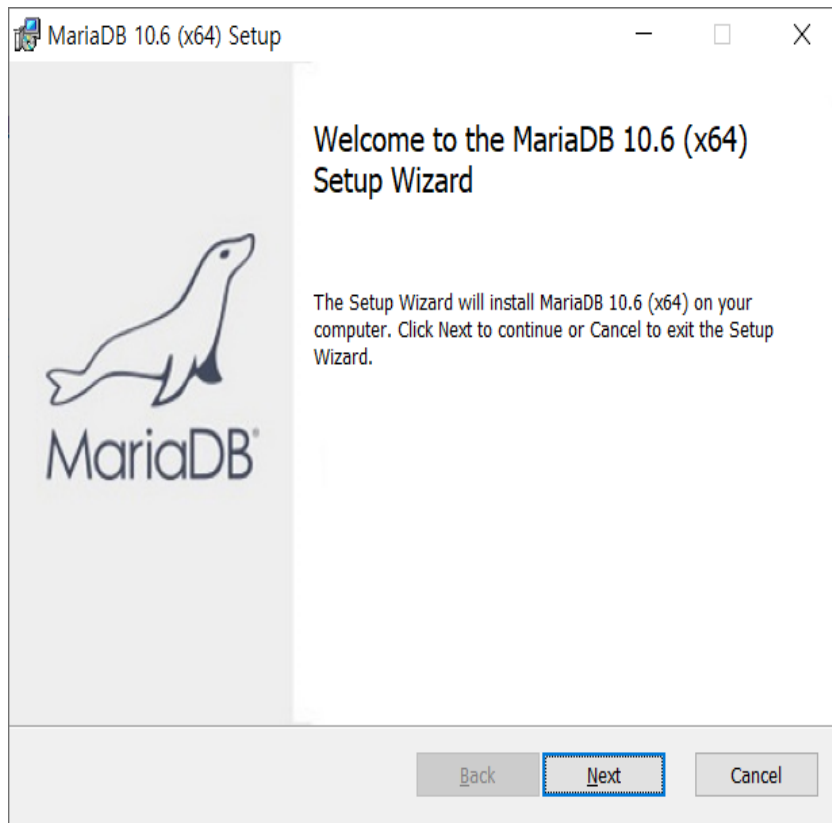
Package Type
MSI Package

Download

Mirror
Blendbyte - Taipei

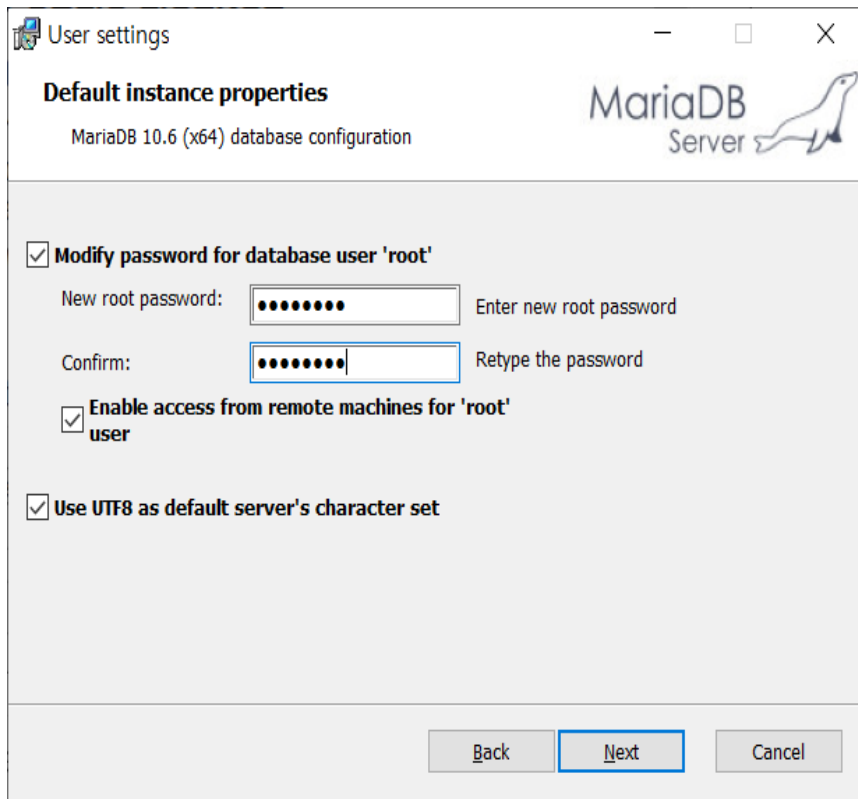
1. Mariadb 설치

☒ 라이선스 동의 조건 체크



1. Mariadb 설치

- ☒ root 암호를 입력하고, 기본 캐릭터셋 UTF8 사용할 수 있게 체크
- ☒ TCP 기본 포트는 3306, 만약 mysql 등 이미 설치가 되어 있으면 3308 등의 다른 포트를 사용



User settings

Default instance properties

MariaDB 10.6 (x64) database configuration

MariaDB Server

☒ Modify password for database user 'root'

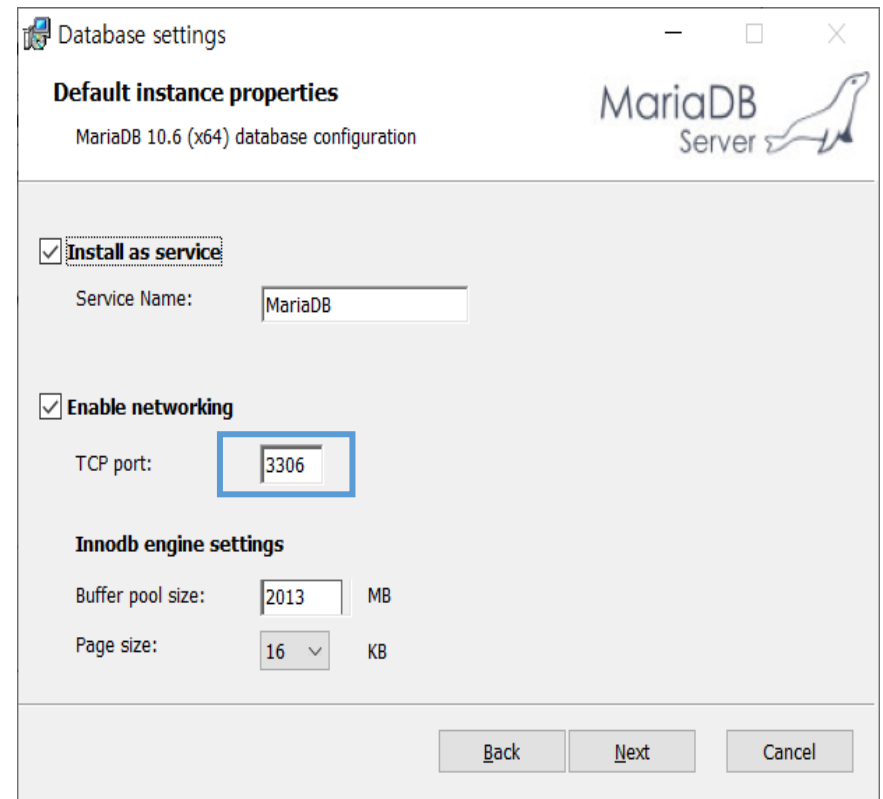
New root password: [password field] Enter new root password

Confirm: [password field] Retype the password

☒ Enable access from remote machines for 'root' user

☒ Use UTF8 as default server's character set

Back Next Cancel



Database settings

Default instance properties

MariaDB 10.6 (x64) database configuration

MariaDB Server

☒ Install as service

Service Name: [MariaDB]

☒ Enable networking

TCP port: [3306]

InnoDB engine settings

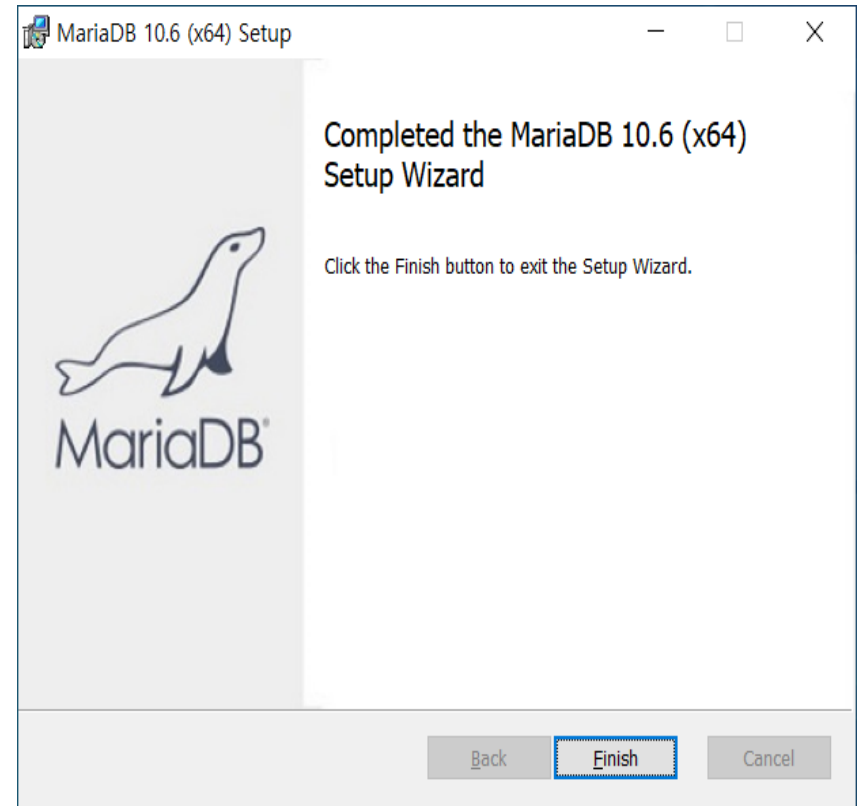
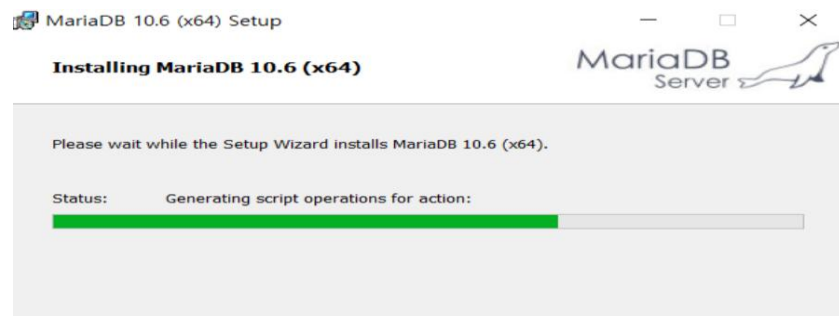
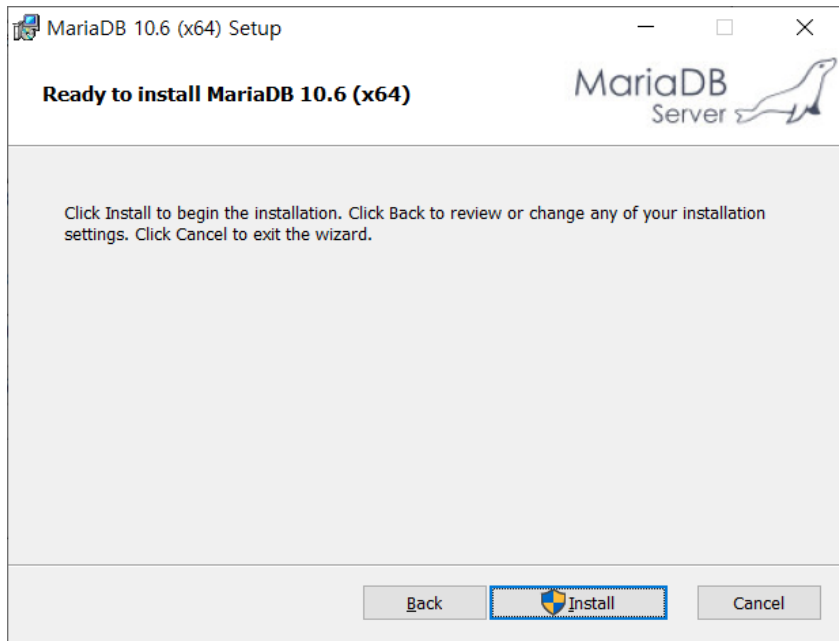
Buffer pool size: [2013] MB

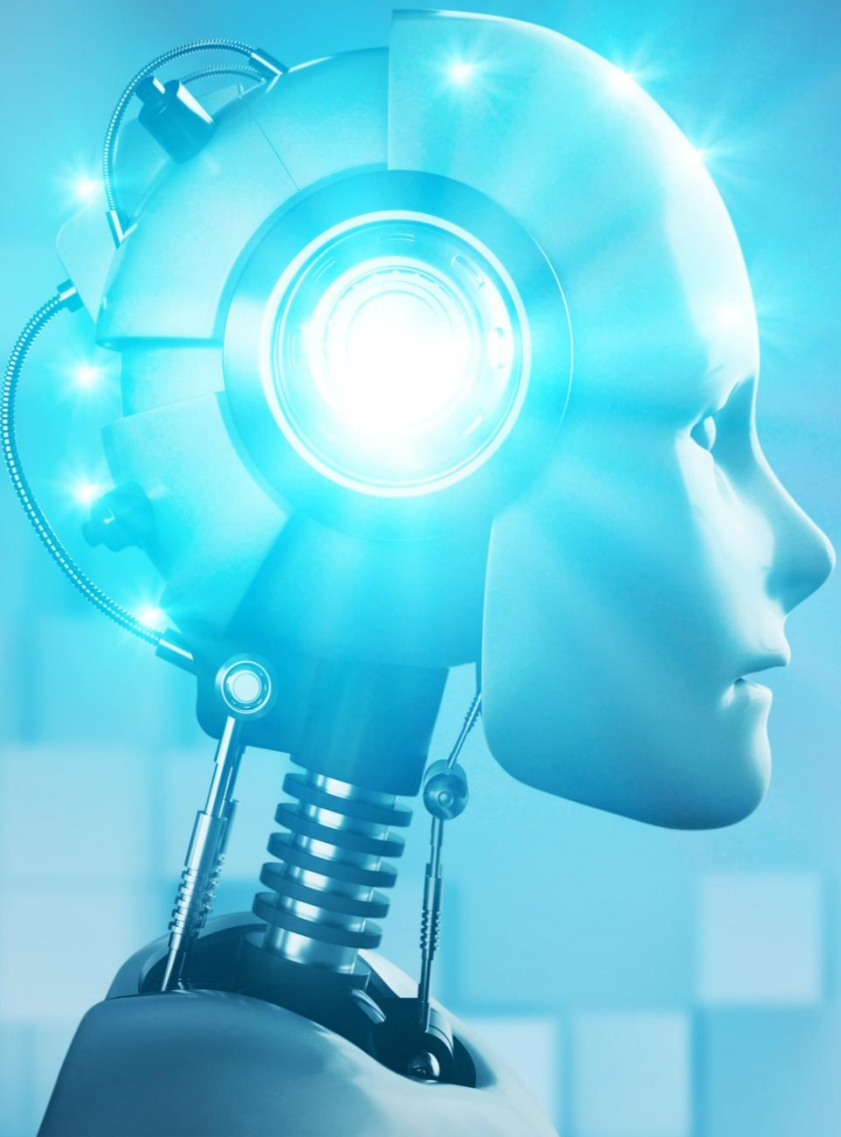
Page size: [16] KB

Back Next Cancel

1. Mariadb 설치

☒ install 버튼 클릭하여 Mariadb 설치



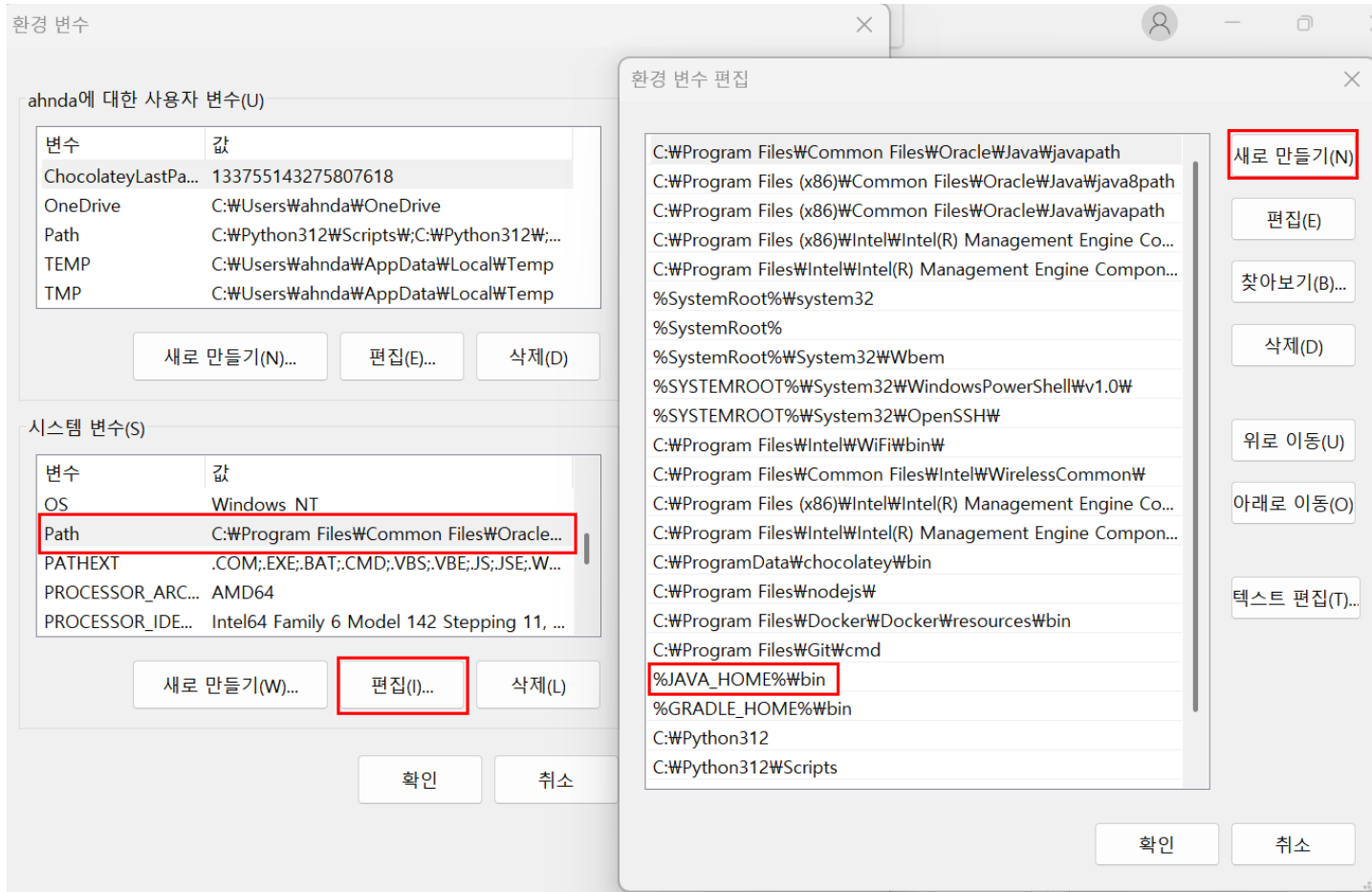


1 DBeaver 설치

1. DBeaver 설치
2. DBeaver 데이터베이스 생성
3. Dbeaver 테이블 생성
4. DBeaver 설정
5. DBeaver 단축키

DBeaver 설치 전에 준비사항

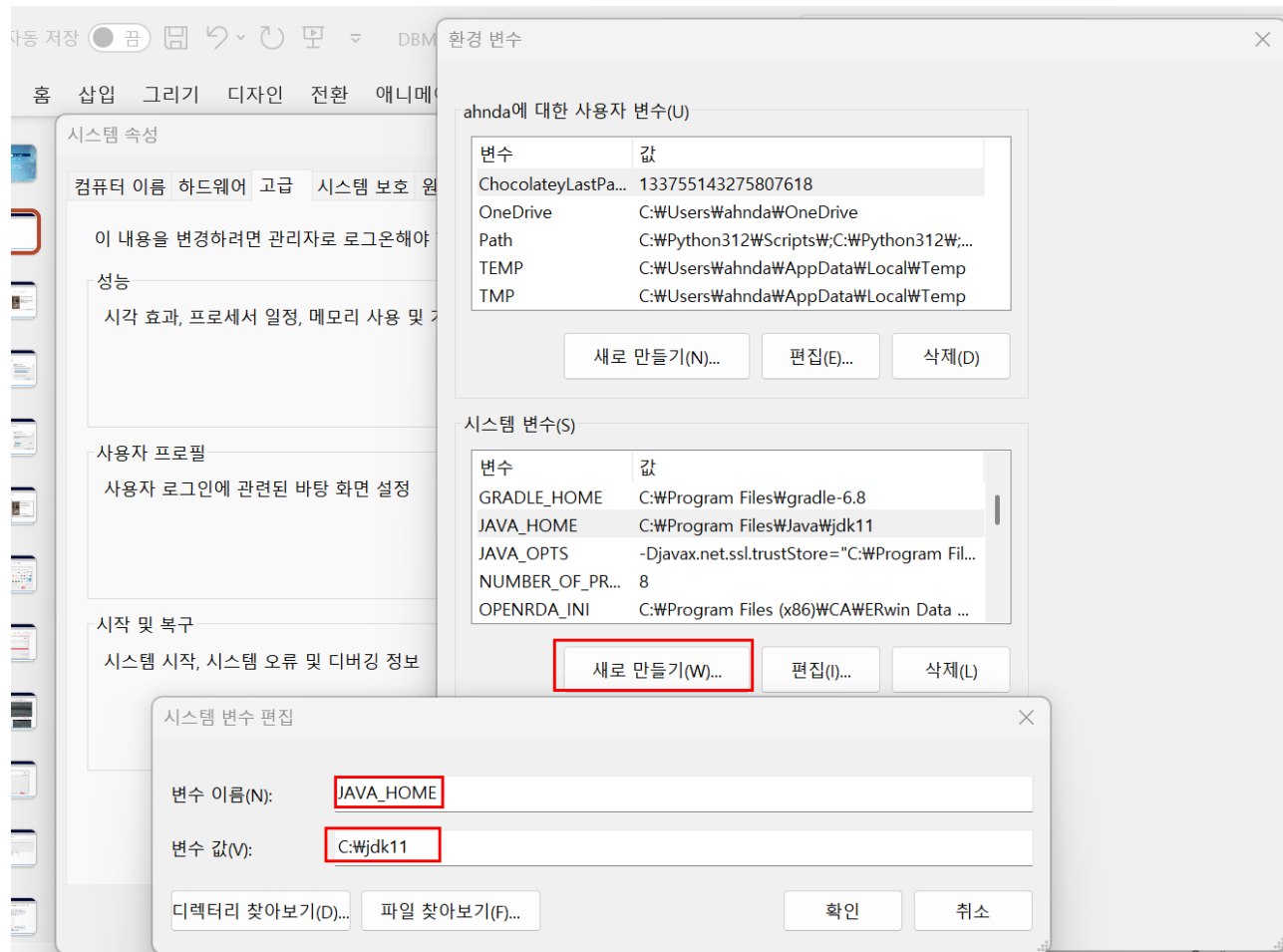
☒ 환경변수 설정 : Path



DBeaver 설치 전에 준비사항

☑ JDK 설치: JDK11 다운로드 (ex. C:\jdk11)

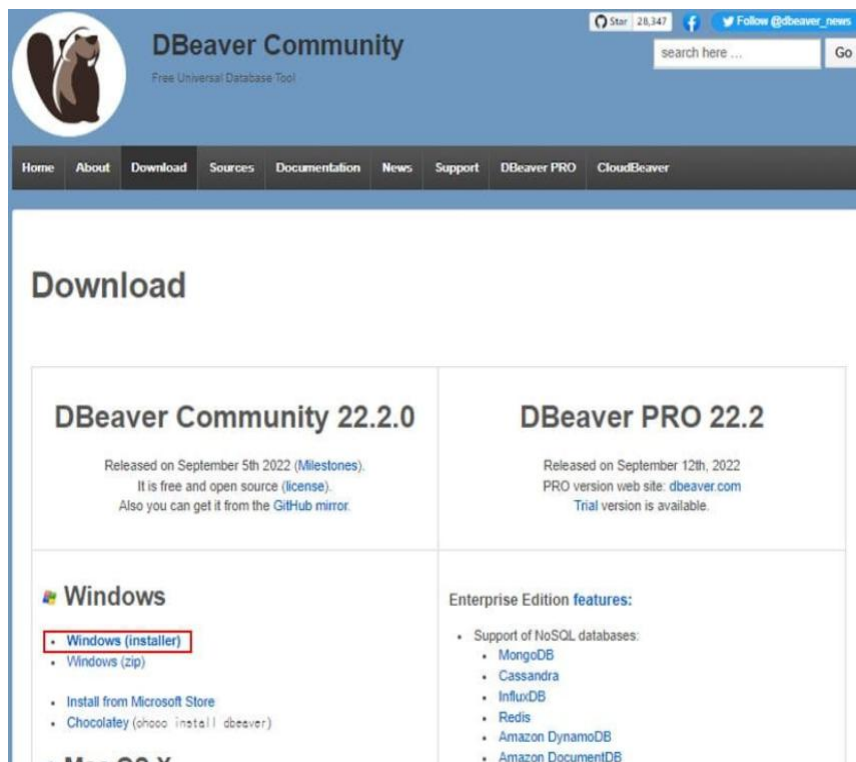
☑ 환경변수 설정



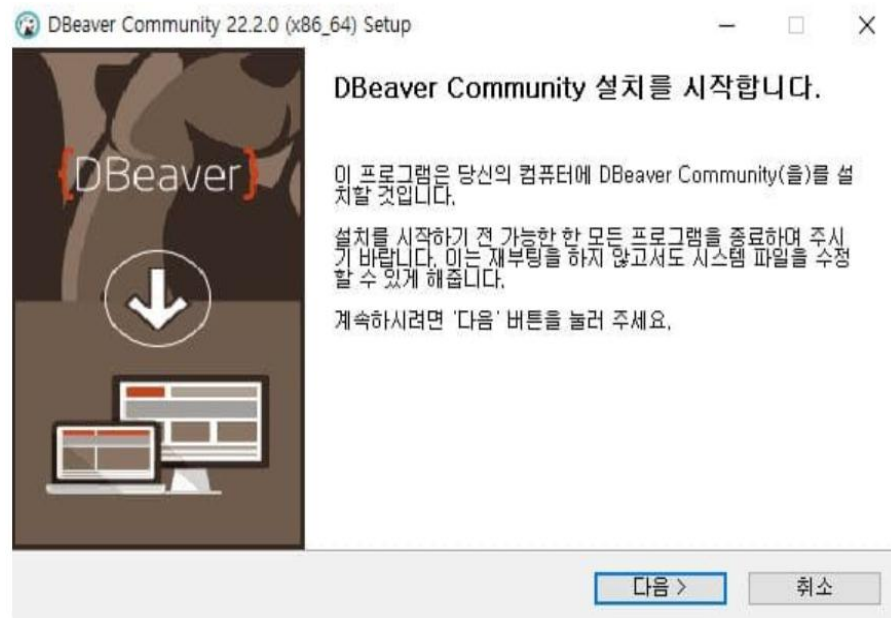
1. DBeaver 설치

☑ 설치 URL : <https://dbeaver.io/download>

☑ DBeaver 설치 : 홈페이지에 나온 버전으로 자유 설치, Windows (installer) 선택 다운로드 및 설치 시작



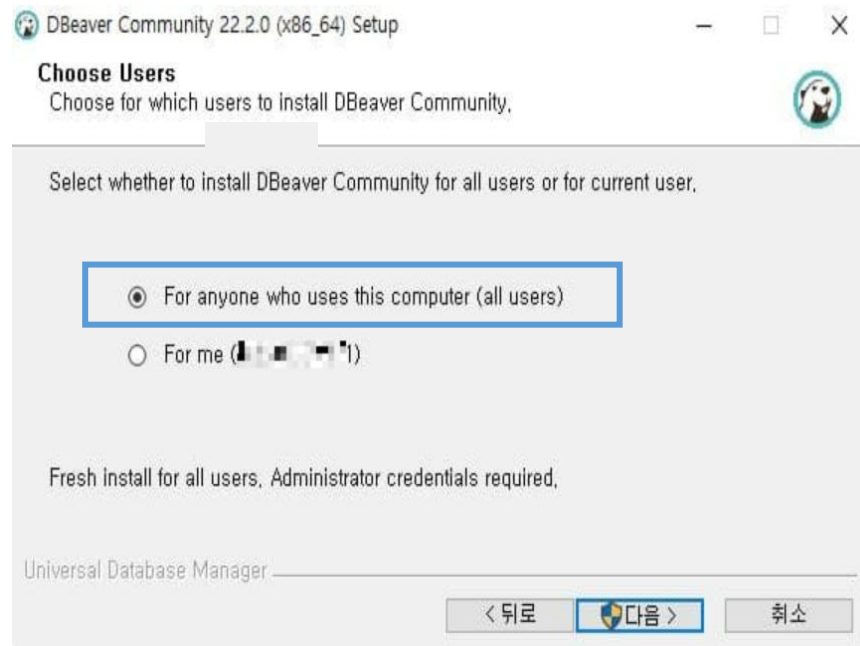
The screenshot shows the DBeaver Community website. The header includes the DBeaver logo, the text 'DBeaver Community Free Universal Database Tool', and social media links. The main navigation bar has links for Home, About, Download, Sources, Documentation, News, Support, DBeaver PRO, and CloudBeaver. The 'Download' section is active, displaying two options: 'DBeaver Community 22.2.0' and 'DBeaver PRO 22.2'. Under 'DBeaver Community 22.2.0', there is a 'Windows' section with a red box around the 'Windows (installer)' link. Other links include 'Windows (zip)', 'Install from Microsoft Store', and 'Chocolatey'. The 'DBeaver PRO 22.2' section lists 'Enterprise Edition features' such as support for NoSQL databases (MongoDB, Cassandra, InfluxDB, Redis, Amazon DynamoDB, Amazon DocumentDB).



The screenshot shows the 'DBeaver Community 22.2.0 (x86_64) Setup' window. The window title is 'DBeaver Community 22.2.0 (x86_64) Setup'. The main content area features the DBeaver logo and a large downward arrow icon. The text reads: 'DBeaver Community 설치를 시작합니다. 이 프로그램은 당신의 컴퓨터에 DBeaver Community(을)를 설치할 것입니다. 설치를 시작하기 전 가능한 한 모든 프로그램을 종료하며 주시기 바랍니다. 이는 재부팅을 하지 않고서도 시스템 파일을 수정할 수 있게 해줍니다. 계속하시려면 '다음' 버튼을 눌러 주세요,'. At the bottom right, there are two buttons: '다음 >' (Next) and '취소' (Cancel).

1. DBeaver 설치

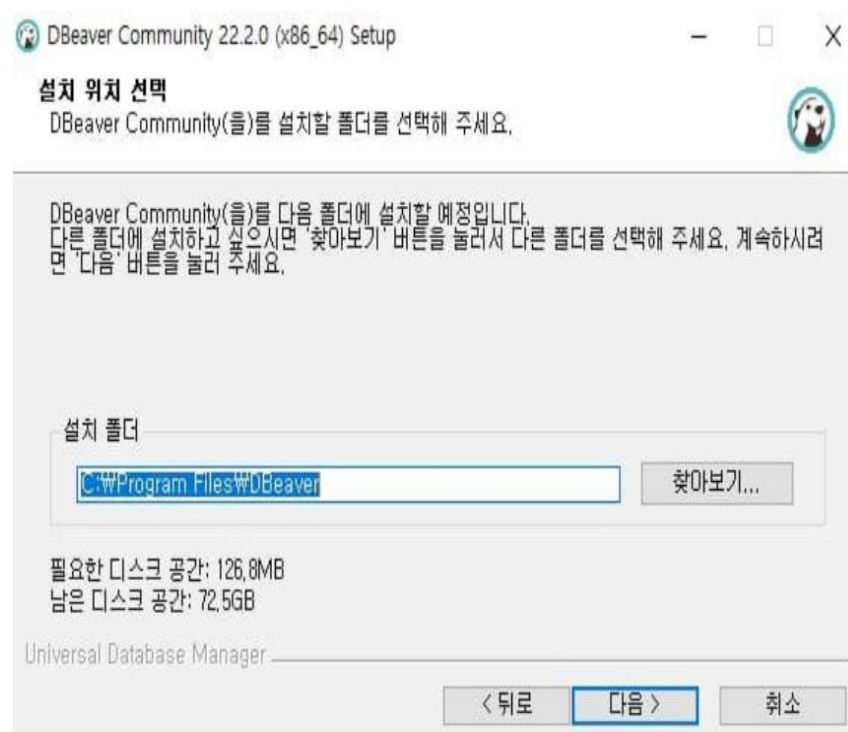
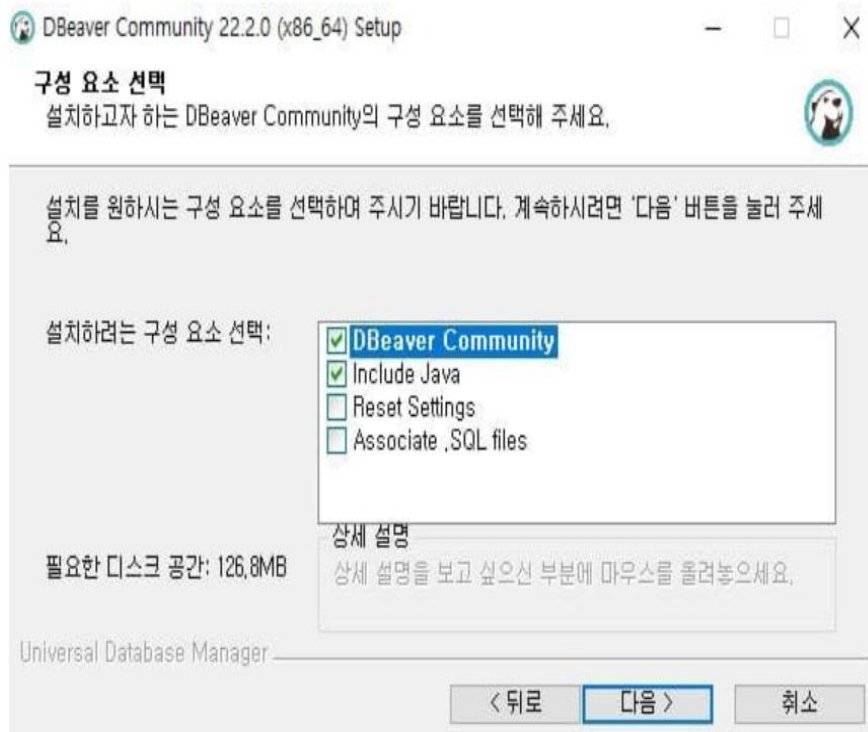
- ☑️ 사용권 계약 동의함 버튼 클릭
- ☑️ For anyone who uses this computer 선택하고 다음버튼 클릭



1. DBeaver 설치

☑ 구성요소 선택 (Dbeaver Community, Include Java)하고 다음 버튼 클릭

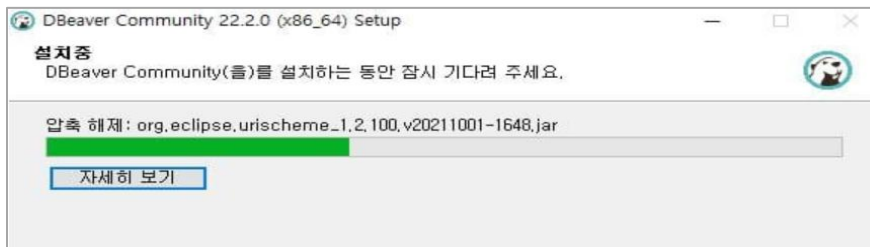
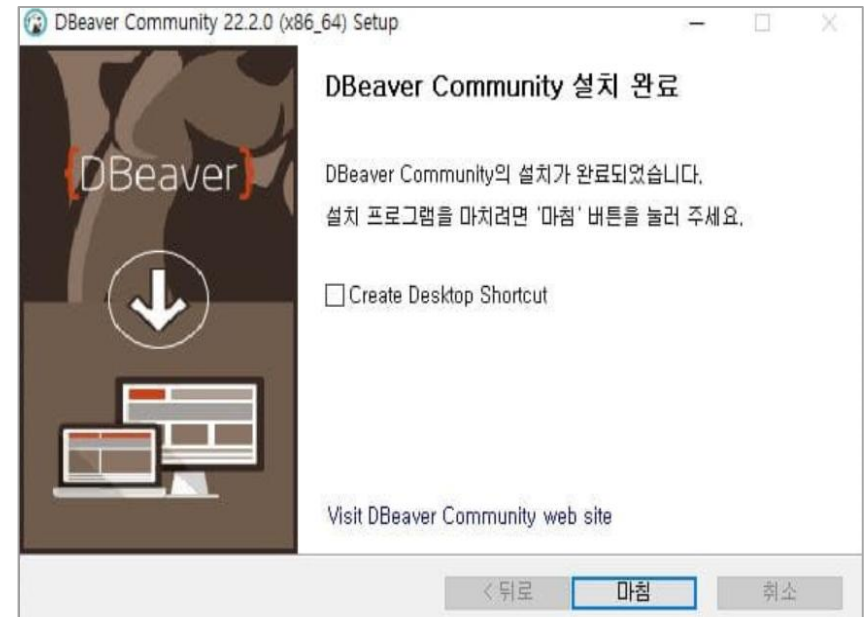
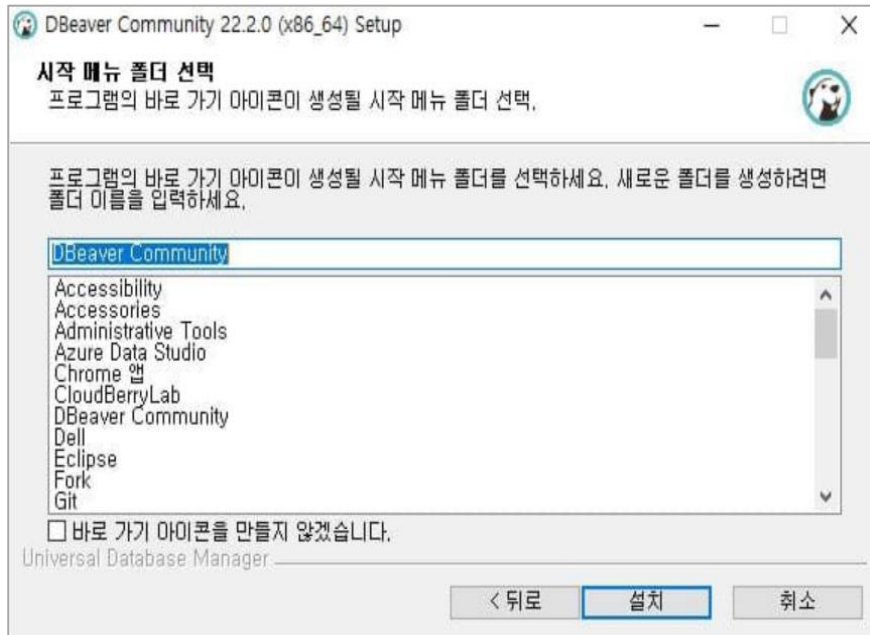
☑ 설치 위치 확인하고 다음 버튼 클릭



1. DBeaver 설치

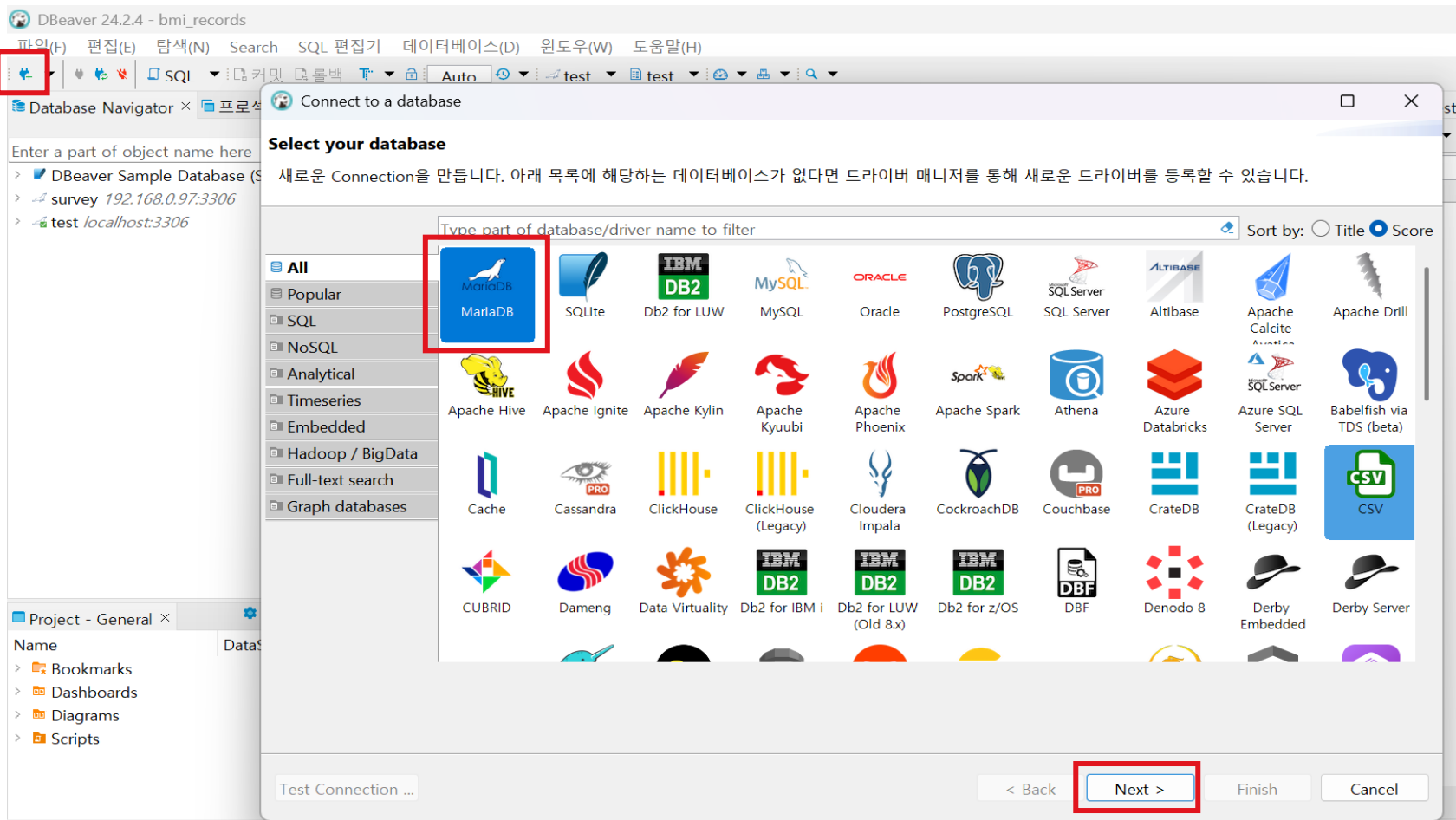
☑ 시작메뉴 폴더 선택(DBeaver Community) 하여 설치 버튼 클릭

☑ 설치완료 마침 버튼 클릭



2. DBeaver 데이터베이스 생성

☑ 좌측 상단 사람모양 파란색 버튼 클릭 → 팝업에서 Mariadb 선택하고 Next 버튼 클릭



2. DBeaver 데이터베이스 생성

- ☑ Host, Port, Database, Username, Password 입력
- ☑ Database 이름 SKIP(이미 생성된 스키마 호출 시에만 사용) , 패스워드는 Mariadb 생성할 때 root에 대한 패스워드 입력
- ☑ Test Connection 버튼 클릭하여 mariadb 드라이버 설치 (※ 반드시 드라이브를 먼저 설치한 후 Finish 버튼 클릭 필요)

The image shows the 'Connect to a database' dialog box in DBeaver, specifically for MariaDB. The 'Main' tab is selected, showing connection settings. A red rectangle highlights the 'Server Host', 'Database', 'Username', and 'Password' fields. The 'Server Host' is set to 'localhost' and the 'Port' is '3306'. The 'Database' field is empty, with a red annotation 'database: 기입 X (신규)' (do not enter database name). The 'Username' is 'root' and the 'Password' is empty, with a red annotation '패스워드 입력' (enter password). The 'Save password' checkbox is checked. Below the highlighted fields, the 'Advanced' section shows 'Server Time Zone' set to 'Auto-detect' and 'Local Client' set to 'MySQL Binaries'. At the bottom, the 'Test Connection ...' button is highlighted with a red rectangle. The 'Finish' button is also visible.

Connect to a database

MariaDB connection settings

Main | Driver properties | SSH | SSL

Server

Connect by: ☒ Host ☐ URL

URL: jdbc:mariadb://localhost:3306/

Server Host: localhost Port: 3306

Database: database: 기입 X (신규)

Authentication (Database Native)

Username: root

Password: 패스워드 입력 ☒ Save password

Advanced

Server Time Zone: Auto-detect

Local Client: MySQL Binaries

Connection variables information

Driver name: MariaDB

Connection details (name, type, ...)

Driver Settings Driver license

Test Connection ...

< Back Next > Finish Cancel

2. DBeaver 데이터베이스 생성

- ☑ Mariadb 처음 연결할 때는 드라이버 설치 화면이 팝업되어 나타나고, Download 버튼 클릭하여 설치
- ☑ 정상 connected 팝업이 뜨면 확인 버튼 클릭

The screenshot shows the DBeaver 'Connect to a database' dialog. The 'Connection Settings' tab is active, showing 'MariaDB connection settings'. The 'Main' sub-tab is selected, displaying fields for 'Server', 'Connect by' (Host selected), 'URL' (jdbc:mariadb://localhost:3306/), 'Server Host' (localhost), 'Database' (database: 기입 X(신규)), 'Authentication (Database Native)' with 'Username' (root) and 'Password' (패스워드 입력), and 'Advanced' settings for 'Server Time Zone' (Auto-detect) and 'Local Client' (MySQL Binaries). A red box highlights the 'Test Connection ...' button at the bottom. A red arrow points from this button to the 'Download driver files' dialog.

The 'Download driver files' dialog is shown, indicating that 'MariaDB driver files are missing' and can be downloaded automatically. It lists the required files:

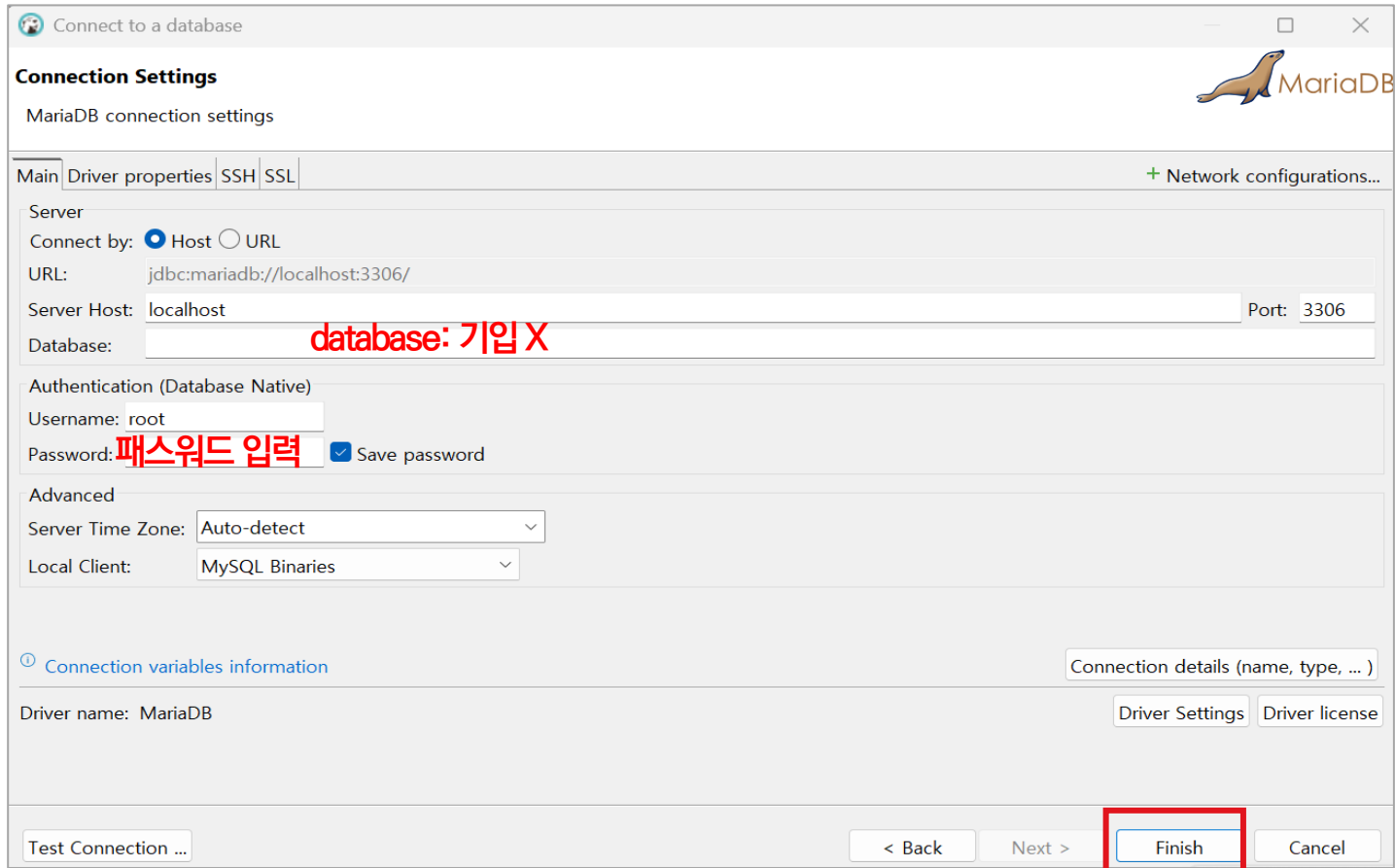
File	Version	Description
org.mariadb.jdbc:mariadb-java-client:RELEASE	3.0.7	JDBC driver for M
https://www.gnu.org/licenses/old-licenses/lgpl-2.1.txt		

Buttons at the bottom include 'Edit Driver', 'Download' (highlighted with a red box), and '취소' (Cancel). A red arrow points from the 'Download' button to the 'Connection test' dialog.

The 'Connection test' dialog shows the result: 'Connected (9504 ms)'. It lists the 'Server' as 'MariaDB 10.10.2-MariaDB-1:10.10.2+maria~ubu2204' and the 'Driver' as 'MariaDB Connector/J 3.0.7'. Buttons at the bottom are '확인' (Confirm) and '세부사항(D) >>' (Details >>).

2. DBeaver 데이터베이스 생성

- ☑ Finish 버튼을 클릭하여 완료



The image shows the 'Connect to a database' dialog box in DBeaver, specifically for MariaDB connection settings. The window has a title bar with standard OS controls and a MariaDB logo in the top right. The main area is titled 'Connection Settings' and contains several sections: 'Main' (selected), 'Driver properties', 'SSH', and 'SSL'. The 'Main' section includes fields for 'Server Host' (localhost), 'Port' (3306), and 'Database' (with a red annotation 'database: 기입 X'). The 'Authentication (Database Native)' section has 'Username' (root) and 'Password' (with a red annotation '패스워드 입력') and a 'Save password' checkbox. The 'Advanced' section has 'Server Time Zone' (Auto-detect) and 'Local Client' (MySQL Binaries). At the bottom, there are buttons for 'Test Connection ...', '< Back', 'Next >', 'Finish' (highlighted with a red box), and 'Cancel'. A 'Connection variables information' link is also present.

Connect to a database

Connection Settings

MariaDB connection settings

Main Driver properties SSH SSL + Network configurations...

Server

Connect by: ☒ Host ☐ URL

URL: jdbc:mariadb://localhost:3306/

Server Host: localhost Port: 3306

Database: database: 기입 X

Authentication (Database Native)

Username: root

Password: 패스워드 입력 ☒ Save password

Advanced

Server Time Zone: Auto-detect

Local Client: MySQL Binaries

① Connection variables information Connection details (name, type, ...)

Driver name: MariaDB Driver Settings Driver license

Test Connection ... < Back Next > Finish Cancel

2. DBeaver 데이터베이스 생성

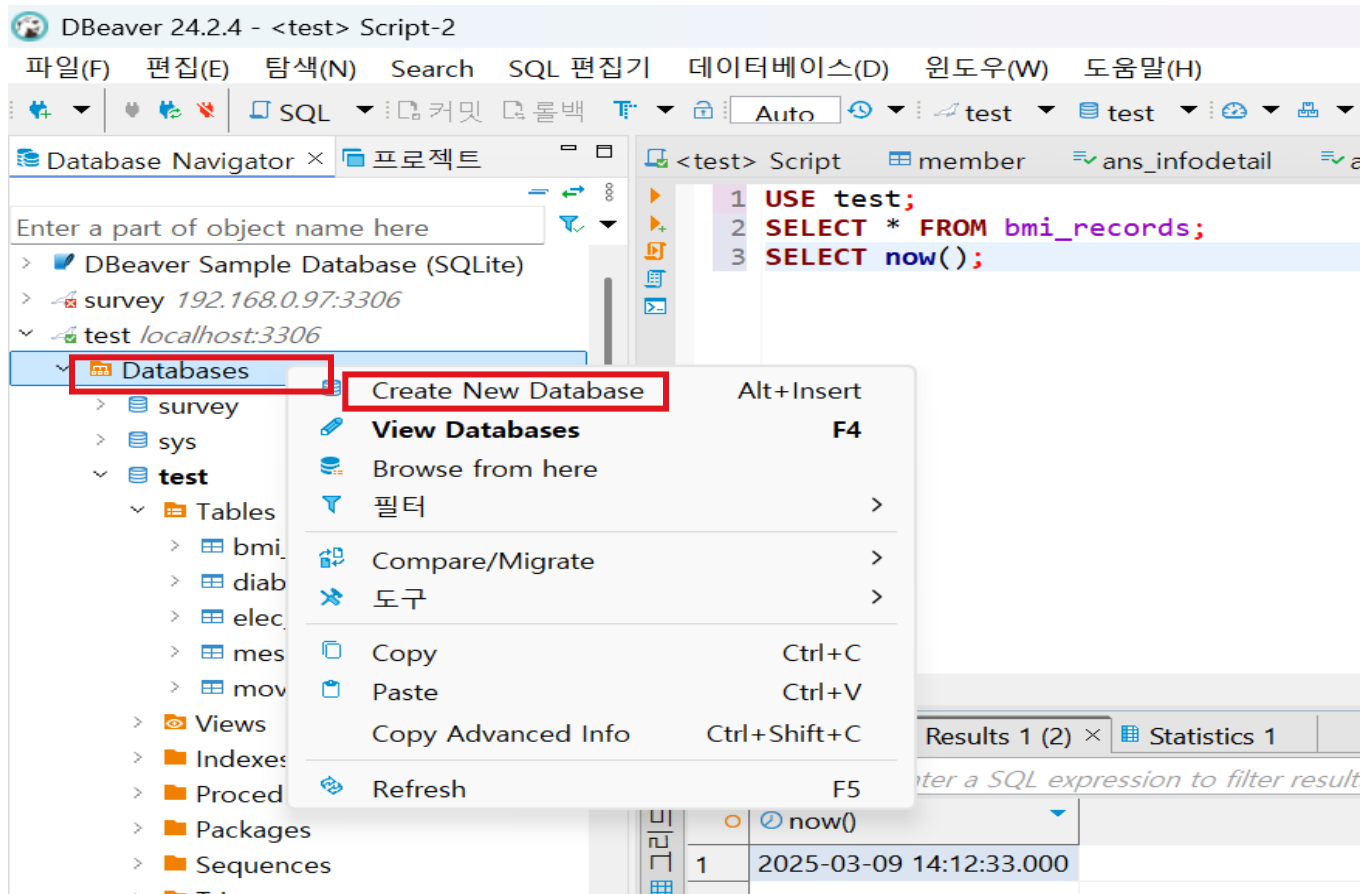
☑ 정상적으로 연결되면 좌측에 설정한 database(ex. test) 확인 가능

The screenshot shows the DBeaver 24.2.4 interface. On the left, the 'Database Navigator' pane displays a tree structure. The 'test' database, located at 'localhost:3306', is highlighted with a red rectangle. The main area on the right shows the 'Properties' tab for the selected 'test' database. The properties include: ID: mariaDB-1930593e4bd-282cf7e2e983bca0, 이름: test, 설명: (empty), Origin: local, 연결 시간: 25. 3. 9. 오후 12:36, and 연결 유형: Development. Below the properties, a table lists the databases in the instance.

	Schema Name	Default Charset	Default Collation	SQL Path	Database size
Databases	survey	utf8mb3	utf8mb3_general		18M
Users	sys	utf8mb3	utf8mb3_general		32K
Administer	test	utf8mb3	utf8mb3_general		17M
System Info					
Driver / ...					

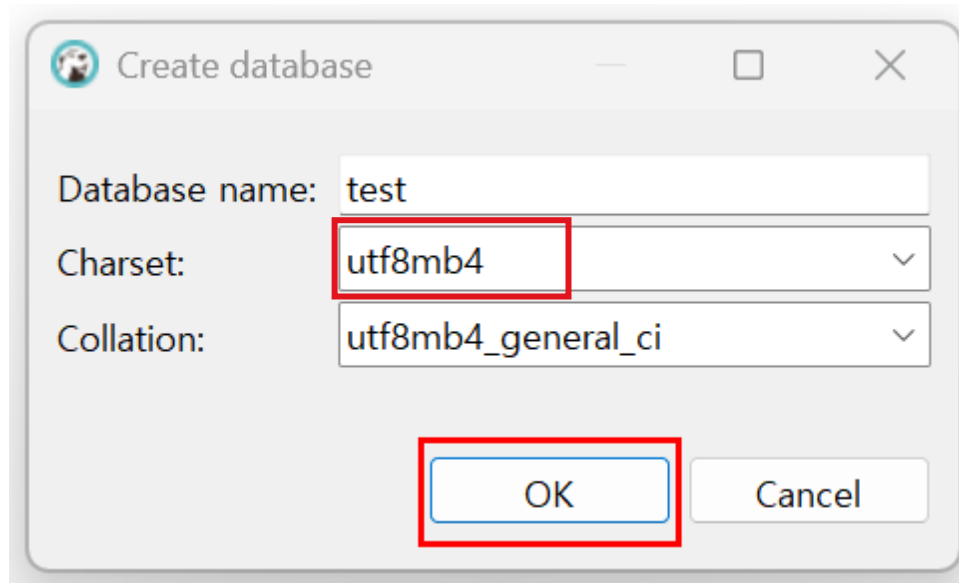
2. DBeaver 데이터베이스 생성

- ☑ 데이터베이스 생성을 위해 생성한 test 스키마 Database에서 오른쪽 클릭해서 Create New Database 선택



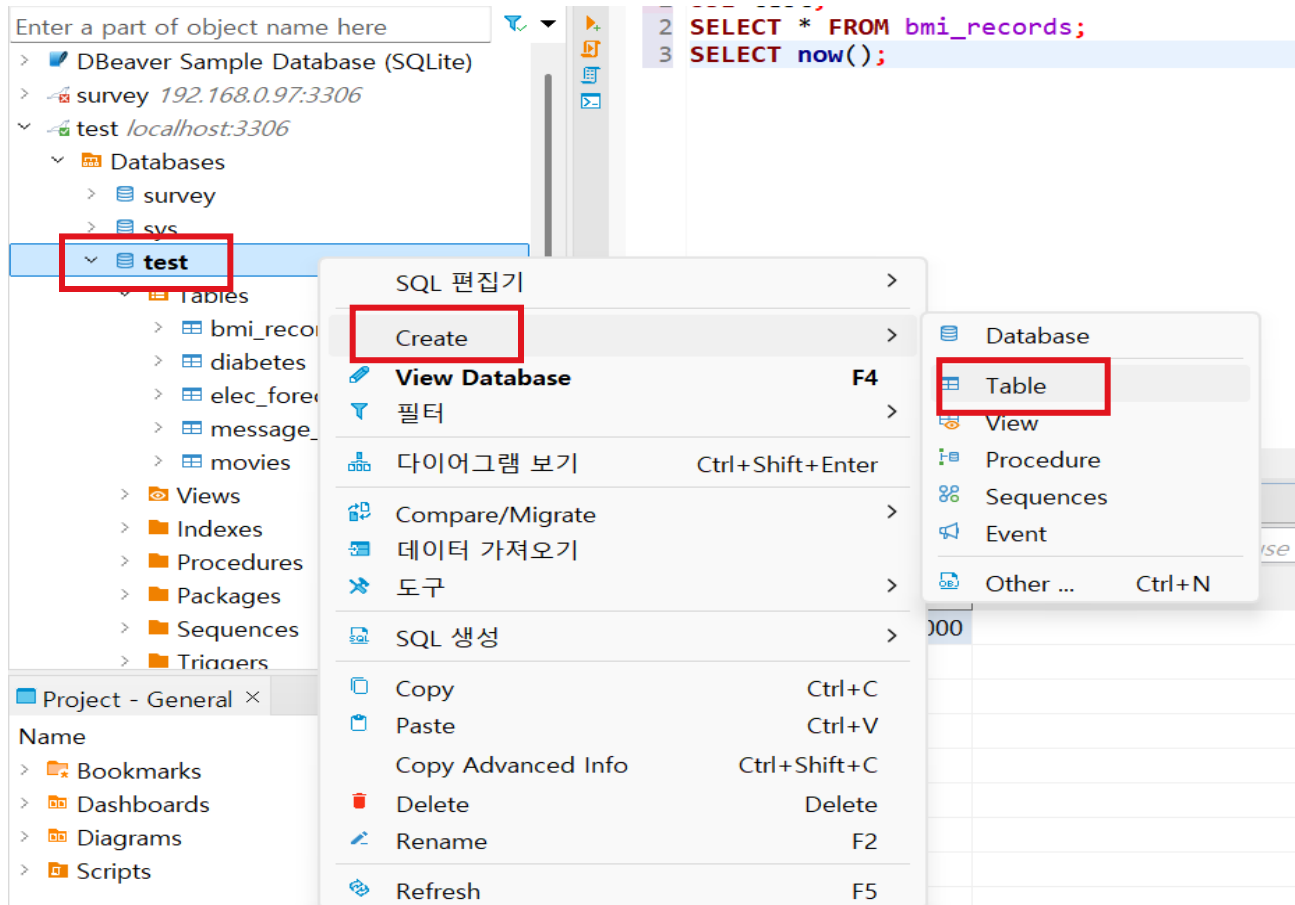
2. DBeaver 데이터베이스 생성

- ☑ Database name을 지정(ex. test)하고, Charset은 utf8mb4 , Collation은 utf8mb4_general_ci 선택하고 ok버튼 클릭
- ☑ utf8mb4 : 4바이트까지 지원하고, 기존 utf8mb3(3바이트)에서 한글, 영문 등 외에도 이모지, 특수문자까지 지원



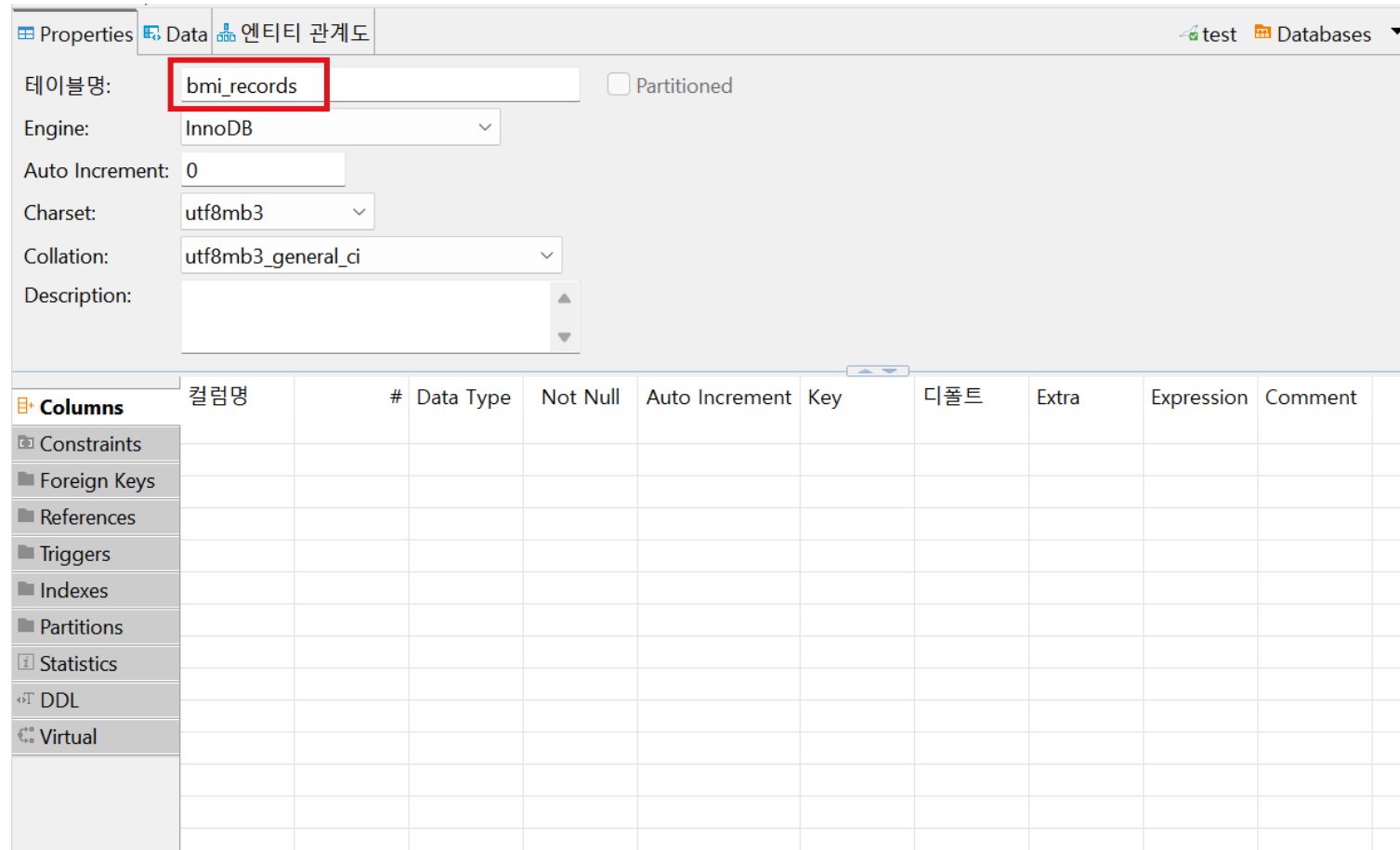
3. DBeaver 테이블 생성

- ☑ 해당 데이터베이스를 선택하여 오른쪽 클릭 후 Create에서 Table 선택



3. DBeaver 테이블 생성

☑ 테이블 명 기입

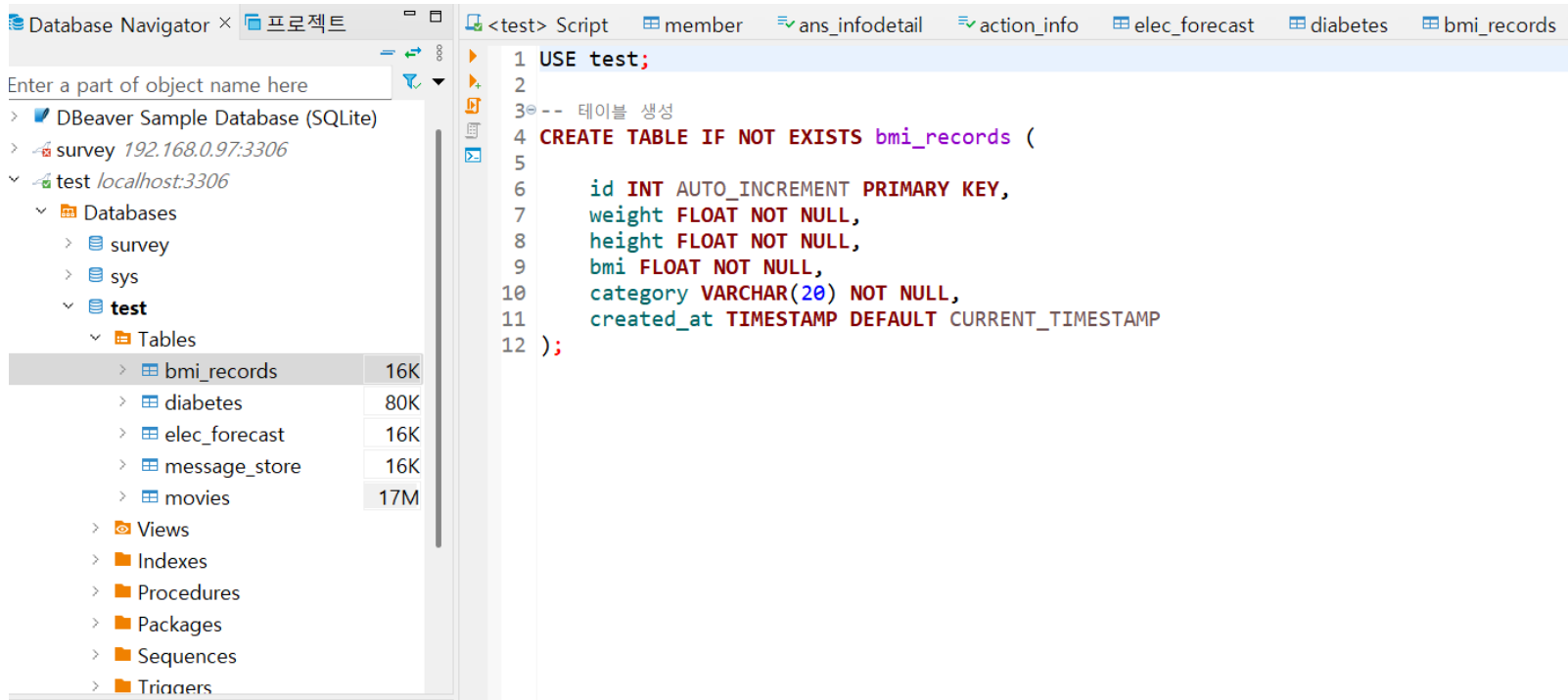


The image shows the DBeaver table creation dialog. The '테이블명' (Table Name) field is highlighted with a red box and contains the text 'bmi_records'. The 'Engine' is set to 'InnoDB', 'Auto Increment' is '0', 'Charset' is 'utf8mb3', and 'Collation' is 'utf8mb3_general_ci'. The 'Partitioned' checkbox is unchecked. Below the properties, there is a table for defining columns and constraints.

	컬럼명	#	Data Type	Not Null	Auto Increment	Key	디폴트	Extra	Expression	Comment
Columns										
Constraints										
Foreign Keys										
References										
Triggers										
Indexes										
Partitions										
Statistics										
DDL										
Virtual										

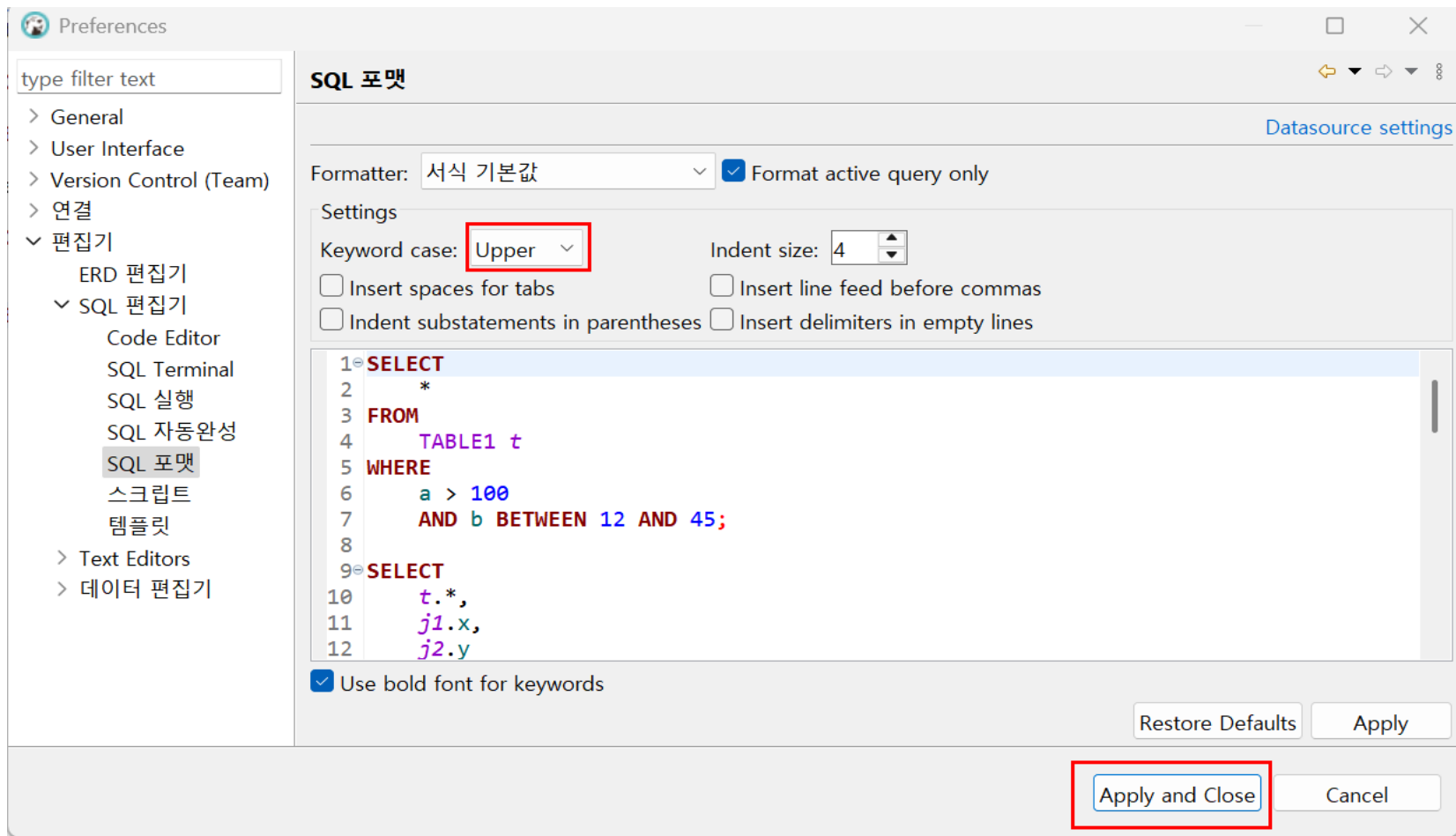
3. DBeaver 테이블 생성

- ❑ 쿼리로 생성하기 위해 상단 메뉴 SQL 편집기 클릭하고 New script 클릭
- ❑ 단축키를 사용할 때에는 CTRL +]
- ❑ SQL 편집기에서 쿼리를 실행할 때는 쿼리를 드래그 한 후 단축 키 CTRL + ENTER



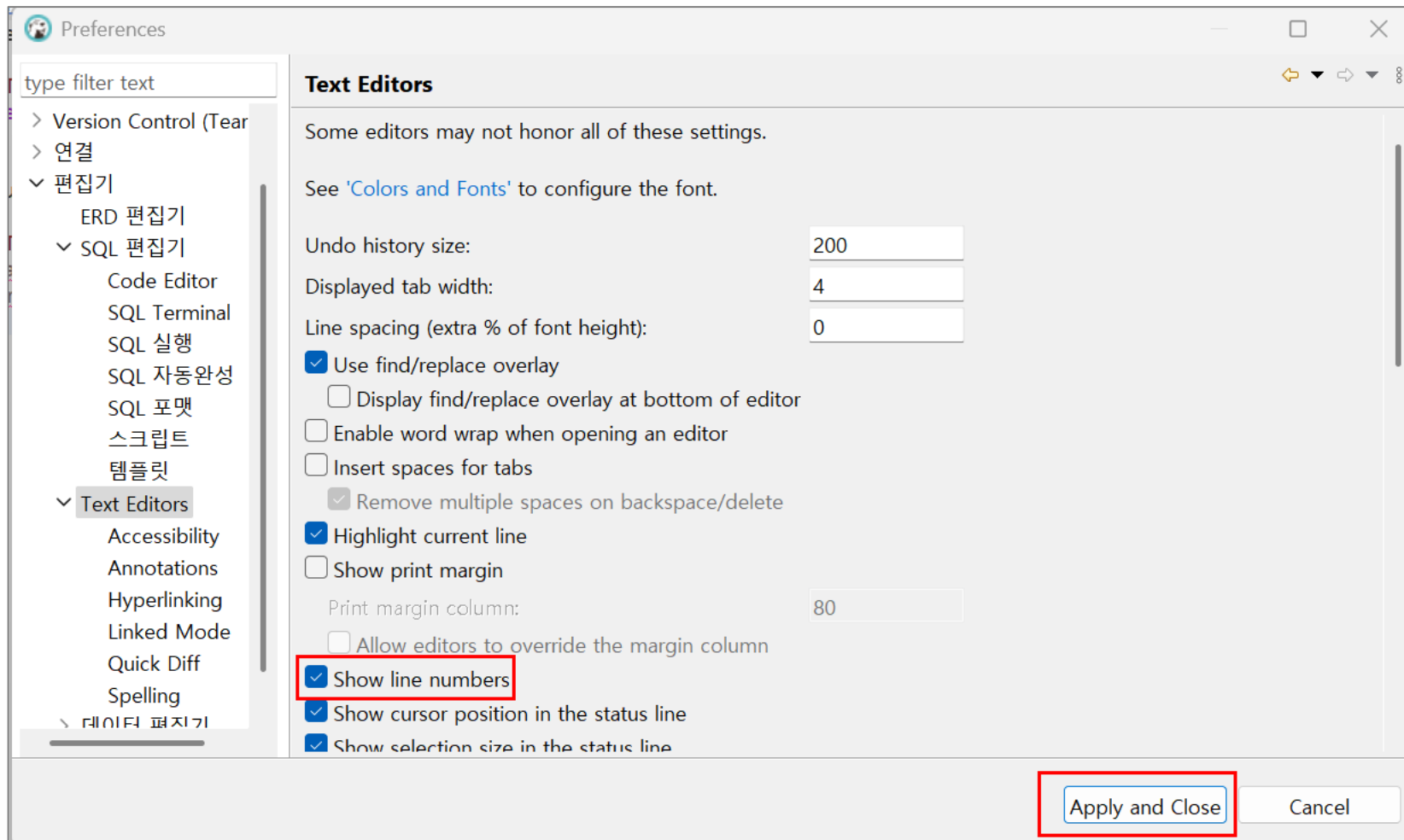
4. DBeaver 설정

- ❑ 쿼리 창에서 select, from, where 등의 주요 키워드가 대문자로 자동 표시되도록 하기 위해, 상단 메뉴 윈도우에서 preferences를 선택하고 편집기에서 SQL 편집기 선택 후 SQL 포맷에서 keyword case를 Upper로 선택하여 저장



4. DBeaver 설정

- ☑ 쿼리 창에서 line 번호가 보이게 하기 위해서 Dbeaver 상단 메뉴에서 윈도우 메뉴 선택하고 Preference 클릭해서 Text Editors에서 show line numbers 선택



5. DBeaver 단축키

☒ Dbeaver에서 자주 쓰는 단축키

단축키	기능
CTRL + ENTER	현재 쿼리 실행
ALT + X	여러 쿼리 한번에 실행
CTRL + \	새 탭에서 현재 쿼리 실행
CTRL + /	현재 단일 행 주석 전환
CTRL + SHIFT + /	여러 행 주석 전환
CTRL +]	새 SQL 편집기 열기
CTRL + SHIFT + F	선택한 SQL문 자동 정리
CTRL + (-)	폰트 줄이기
CTRL + (+)	폰트 키우기